

BSCHE Graduate Written Examination – Fall 2019

Written examination conducted with graduating seniors in Chemical Engineering

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Question 1

Career plans. (Scoring rubric was designed to assess potential for life-long learning.)

Where do you see yourself in 5-10 years, and what steps will you take to get there?

Probably working in a chemistry related field.

I'm networking with multiple companies that deal with chemicals, especially if the job scope requires organic chemistry.

How have you researched your career plans and options (companies, graduate schools, PE certification, etc.)?

Yes.

Does your prospective employer sponsor master's degree or MBA? Under which terms?

No.

Question 2

Professional ethics and ethical ramifications of technology.

Can you name a recent news item or significant piece of legislation that impacts on technology or engineering, and comment on its ethical ramifications?

N/A

Have you experienced an ethical dilemma in your educational or work (e.g., co-op) experience?

N/A

How would you approach an ethical dilemma? Where or to whom would you turn for guidance? Can ethical problems be approached in terms of a systematic methodology, or are you guided more by gut reaction, personal experience, upbringing, etc.?

N/A

Are you familiar with the AIChE Code of Ethics? Can you name specific provisions in it?

1. Hold paramount the safety, health and welfare of the public and protect the environment in performance of their professional duties.
2. Formally advise their employers or clients (and consider further disclosure, if warranted) if they perceive that a consequence of their duties will adversely affect the present or future health or safety of their colleagues or the public.
3. Accept responsibility for their actions, seek and heed critical review of their work and offer objective criticism of the work of others.

4. Issue statements or present information only in an objective and truthful manner.
5. Act in professional matters for each employer or client as faithful agents or trustees, avoiding conflicts of interest and never breaching confidentiality.
6. Treat all colleagues and co-workers fairly and respectfully, recognizing their unique contributions and capabilities by fostering an environment of equity, diversity and inclusion.
7. Perform professional services only in areas of their competence.
8. Build their professional reputations on the merits of their services.
9. Continue their professional development throughout their careers, and provide opportunities for the professional development of those under their supervision.
10. Never tolerate harassment.
11. Conduct themselves in a fair, honorable and respectful manner.

Which courses at UIC have contributed most to your views on professional ethics?
How?

~~All courses~~ CHE396: Senior Design I. The assignment on ethics allowed me to apply the code of conducts.

Question 3

Working in multidisciplinary teams

Can you name two other types of engineers that you might find yourself working with?

Mechanical, Biomedical

In what context or application would you be collaborating with them?

Mechanical engineers work with plant machinery

Question 4

Numerical methods and computational proficiency (Excel, MATLAB or VBA): choose one problem.

(a) Generate a graph of the solution of the following ODE initial value problem.

$$\frac{dx}{dt} = \frac{1}{\tau} [x_{\text{in}}(t) - x(t)] - kx(t)^2, \quad x(0) = 2$$
$$x_{\text{in}}(t) = 1 + \varepsilon \sin(\omega t)$$

with the constants $\tau = 3.4$, $k = 1.2$, $\varepsilon = 0.1$ and $\omega = 5$. Can you think of a chemical engineering problem (or, at least, a course) in which such an ODE would arise?

→ Reaction Engineering

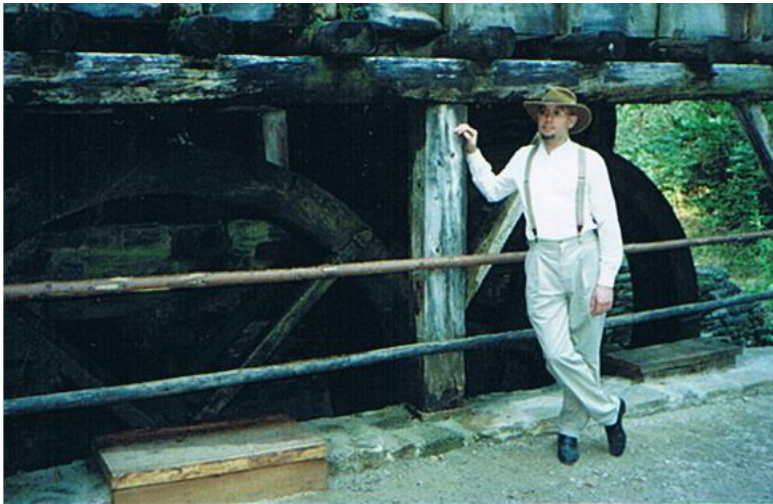
(b) Find a numerical value for the following definite integral.

$$\int_1^7 \frac{dx}{1 + \cos^3 x + \ln x} = \boxed{3.156}$$

Question 5

Engineering estimate.

The waterwheel shown in the following three pictures was observed to make one complete revolution every 15 seconds. Estimate the power output (in units of horsepower)?



$$P = 7000 \dot{q} h$$

$$15 \text{ sec} = 1 \text{ revolution}$$

$$\dot{q} = v \cdot A$$

$$v = \left(\frac{1 \text{ rev}}{15 \text{ sec}} \right) \left(\frac{2\pi \text{ rad}}{1 \text{ rev}} \right) \left(\frac{1.22 \text{ m}}{1 \text{ rad}} \right) = 0.511 \text{ m/s}$$

$$h_v = \frac{v^2}{2g} = \frac{0.511}{19.62} = 0.0133 \text{ m}$$

$$h_p = 3.5 \text{ m (assumed)}$$

$$\therefore h = 3.5 + 0.0133 = 3.5133 \text{ m}$$

$$\dot{q} = (0.511)(0.3716) = 0.19 \text{ m}^3/\text{s}$$

$$\therefore P = 7000(0.19)(3.5133) = 4672.69 \text{ W}$$

$$= \boxed{6.3 \text{ HP}} \quad \#$$

P: Power (W)

$\dot{q}$: vol. flow rate (m^3/s)

h: Head (m)

* Assume radius = 1.22 m)